INDEX

NAME: Yaswanth Sreenivas STD.: ____ SEC.: ____ ROLL NO.: ____ SUB.: CN observation

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Study of Network commands

Window commands :-

1. arp -a – ARP is short form of address resolution protocol

O/P:-

```
Interface : 10.16.56.150 ..... 0x9
  Internet address      Physical address      Type
  10.16.56.155          ec-12-17-17-12-5c     dynamic
  10.16.56.255          ff-ff-ff-ff-ff-ff     static
```

2. hostname: This is the simplest of all TCP/IP command. It will display the name of your computer

O/P:- LAPTOP-T004TKTN

3. ipconfig /all :- This command displays configuration information about your TCP/IP connection

O/P:-

Window IP config

Hostname : LAPTOP-T004TKTN

Node type : Mixed

4. nbstat -a : This command helps solve problems with NetBIOS name resolution

O/P :- Ethernet 2:

Node IP Address : [192.168.56.1] Aliupe IP:[]

Host not found

5. netstat : (Network Statics) netstat display a variety of statistics about a computer Active TCP/Ip connection.

O/P : Active connection

Proto	local Address	Foreign Address
TCP	10.16.56.50.944	4.213.25.242

6. nslookup : It is used to diplay the DNS details such as Ip address of a particular computer.

O/P : Server : Unknown

Address : 10.16.56.155

7. pathping : Pathping in a unique to window and its basically a combination of the ping and Tracert Commands

Cmd :- pathping — g google.com

O/P :

Tracing route to google.com [142.35.41.14]

Trace complete

8. Ping : (Packet Internet Groper) Command is the bat way to tat connectivity between two nodes.

Ping google.com

O/P :

Pinging

Linux Commands :

1. Ip Commands

1. Cmd : ip address show - Displays all IP addresses

O/P : etho : inet 192.73.65.64.12401

2. ip address add 192.168.1-254/24 dev enp0s3 - ass igns ip address to interfaces

O/P ; IP address added successfully

3. IP link ect etho ip — Enables the network interface.

O/P : etho state up

4. ip link set etho down - disables the network interface

O/P : etho state Down

5. ip link set etho promic on - Enable prominuous mode

O/P : Promicuous mode enabled

6. Zp route add default via 192.168.1.254 dev etho - Adds the default gate way

O/P : 10.10.1.4 via 192.168.1.254 dev eth0

2. ifconfig :-

Displays network interface configuration

O/P :-

eth0 link encap : Ethernet inet 192.168.1.10

3. mtr :-

1. mtr google.com - shows route statistics to host.

O/P : 142.250.183.206.

Host : Localhost

2. mtr -g google.com - shows numeric IP address only.

O/P : google.com (142.250.183.206)

3. mtr -b google.com - shows both host name and IP address.

O/P : google.com (142.250.183.206)

4. mtr -c 10 google.com - sends only 10 probe products.

O/P : Sent 10 packets

3 tcp dump

1. dnf install -y tcdump - Install tcpdump

3. tcpdump -i eth0 -c 10 - capture packets to/from a host

O/P : 10 packets captured

4. tcp dump -i eth0 src host 8.8.8.8 - Captures packets from a host

O/P : 8.8.8.8 -> 192.168.1.10

5. tcp dump -i eth0 dst host 8.8.8.8 captures packets to a host.

O/P : 192.168.1.10 > 8.8.8.8

6. tcpdump -i eth0 port 53 - Captures DNS traffic

O/P : DNS Query captured

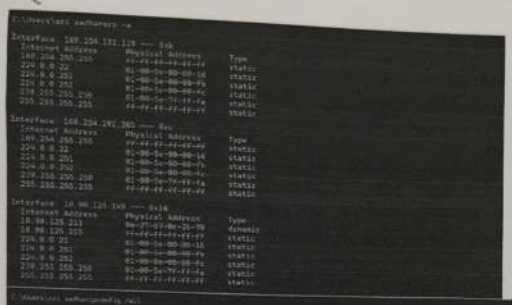
7. tcpdump -i eth0 -c 10 host www.google.com and port 443 - Captures HTTPS products.

O/P : TCP 443 captured

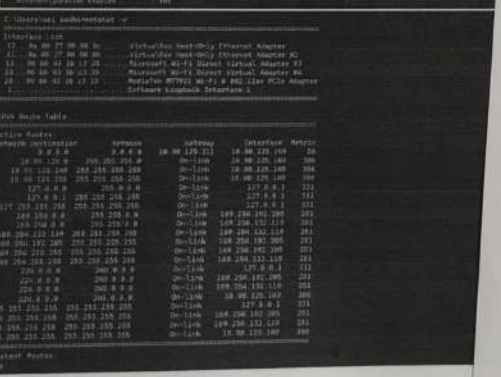
8. tcp dump -i eth0 port not 53 and not 25 - Captures all traffic except DNS and SMTP.

O/P : TCP/IP packets displayed

5. Ping.







5. nmcli connection modify "wired connection 1"
ipv6.method auto — configures automatic
IPV6.

O/P: IPV6 method set to auto

6 nmcli connection up "Internal - Lan" - Activates
the connection profile

Result:

The above both the windows and Linux commands are executed and outputs are completed successfully

3/8/26.

Exp:2

Types of Network Cables

Aim: To study of different types of network cable

a) Understand different types of network cables.

Cable Type	Category	Max data rate	Adv & direct	Application
i) UTP	3	10 Mbps	i) cheapest ii) More prone to EMI	10-base T-ethernet
ii) STP	6/6a	100 Mbps	i) Shielded ii) expensive	Used in databases
iii) SSTP	7	10 Gbps	i) Greater effort initials	10 Gb ethernet
iv) Coaxial cable	RG-6 RG-59	10-100 Mbps	i) High bandwidth ii) Size in 6	TV network
v) Fibre optics cable.	Single mode	100 Gbps	i) High Speed ii) Expensive	Max dist is 100m

b) Make your own ethernet cross over cable

Step 1:- To start construction of device, begin by threading shield onto cable.

Step 2:- Next, strip approximately 1 inch of cable. Shielding from both ends. The crimping tool bar round area.

Step 3:- After you need to untangle the wire three should be four twisted pairs.

Step 4:- Once the order is correct, bunch them together in line, and if there are any that stick out further than the snip them back to create an even end.

Step 5:- Next push the cable right in the right end of plug needs to jump over the cable shielding.

Step 6:- After the wires are securely sitting inside the plug, insert it into the crimping tool and push low.

Step 7:- Lastly, repeat for other end using diagram B.

Result: The different type of network cable is studied.

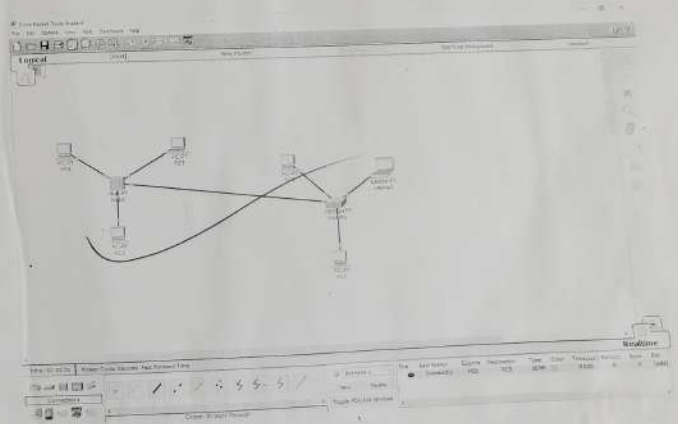
Ex: 3

PACKET TRACER TOOL

Aim: To study the packet tracer tool installation and user interface overview.

Procedure:

1. From the network component box, click and drag and drop the below components,
   a, 4 Generic PCs and One Hub
   b, 4 Generic PCs and One switch
2. Click on Connections:
   * Click on copper-straight-through cable,
   * Select one of the PC and connect it to HUB using cable. The link LED should glow in green. indicating that the link is up. Similarly connect remaining 3 pcs to the HUB.
   * Similarly connect 4 PCs to the Switch using copper straight-through cable.
3. Click on the PCs connected to hub, go to the Desktop tab, click on IP Config, and enter an IP address an IP address and subnet mask. Here, the default gateway and DNS server info is not needed as there are only two end devices in the network click on the PEN (message icon) from the common tool bar,
   a, Drag and drop it on one of PC (source machine) and then drop it on another PC
   → Observe the flow of PDU from source PC to destination PC by selecting the Realtime mode of simulation
   → Repeat step 3 to step 5 for the PCs connected to the switch
   → Observe how HUB and switch are forwarding the PDU and write your observation and conclusion about the behaviours of switch and HUB





Result: Thus the installation and overview has been successfully completed.

Exp 4

Setting Up a Local Area Network

Aim: To design, setup and configure a Local Area Network (LAN) connect 4 computers through a network switch, assign static IP addresses to all devices, verify connectivity using the ping command, and demonstrate resource/data sharing across the network.

Network Diagram:



Theory: The Local Area Network (LAN) is a network that connects computers and devices within a limited network geometrical area.

IP Config Table:

Device	IP address	Subnet Mask	Connected to
PC0	10.1.1.1	255.0.0.0	Switch port Fa 0/1
PC1	10.1.1.2	255.00.0	Switch port Fa 0/2
PC2	10.1.1.3	255.0.0.0	Switch port Fa 0/3
PC3	10.1.1.4	255.0.0.0	Switch port Fa 0/4
Switch	10.1.1.5	255.0.0.0	Switch port Fa 0/5

Implementation in Cisco Packet Tracer:

Since physical hardware was limited, the LAN was also implemented and verified, using cisco packet tracer simulation software, following the below steps:-

Step 1 - Place the Devices

* Open cisco packet tracer
* From the device category panel, select end devices and drag 4 PCs onto the workspace.
* Select switches and drag 1 switch (2960 series) onto the workspace.

Step 2 - Cable the Devices

* Click connection → select copper straight through cable.
* Click each PC, then click the switch and select an available Fast Ethernet port to complete each connection.
* Repeat all for all four PCs (Ports Fa0/1 to Fa0/4). Green Lights confirm a successfull physical connection.

Step 3 - Assign IP addresses to PCs

* Click PC → Desktop tab → IP config
* Enter the static IP Address and Subnet mask as per the IP table (Section C) for each PC

Step 4 - Configure the switch Mangement IP (optional)

* Click the switch → cli tab
* Enter the following commands to assign ca management

IP

enable
configure terminal
interface vlan 1
ip address 10.1.1.5 255.0.0.0
no shutdown
exit

Step 5 - Verify connectivity using Ping.

* Click on PC → Desktop tab → Command prompt
* Use the ping command to test connectivity to the other PCs, eg - ping 10.1.1.2
* Successful "Reply from..." messages confirm Connectivity

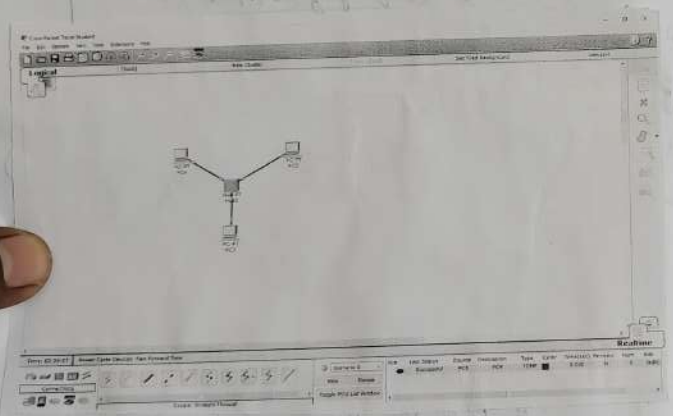
Step 6 - Demonstrate Data Sharing (Server Simulation)

* Add a server device (End devices → Server-PT) and connect it to the switch
* Assign it an IP address (eg., 10.1.1.10, Subnet mask 255.0.0.0) via its desktop → IP config tasks.
* Open per Server's services (config) tab & enable HTTP service.

Step 7: Observes Packet flow in Simulation

* Switch to simulation mode (bottom right of the workspace)

* Send a ping or web req btw hosts and use the play/capture controls to step through the packet as it travels. Hop by hop through the switch observing MAC addresses based forwarding



Results: We Successfully designed, setup and configured a Local Area Network (LAN).

Exp 5

PACKET CAPTURE TOOLS: WIRE SHARP

AIM:

The experiment on Packet capture tool wiresprak.

Description: Wireshark, a network analysis tool formerly known as Ethereal, captures packets in real time and display them in human readable format

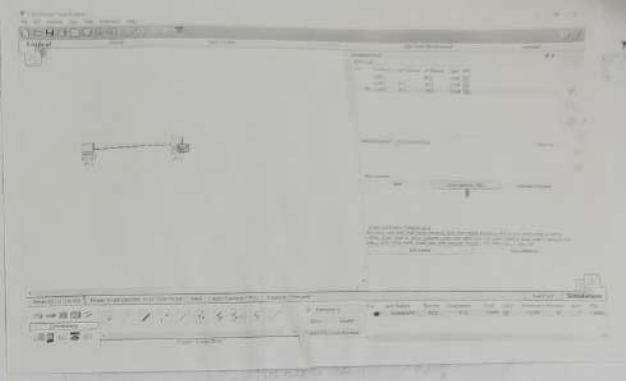
Capturing Packets:

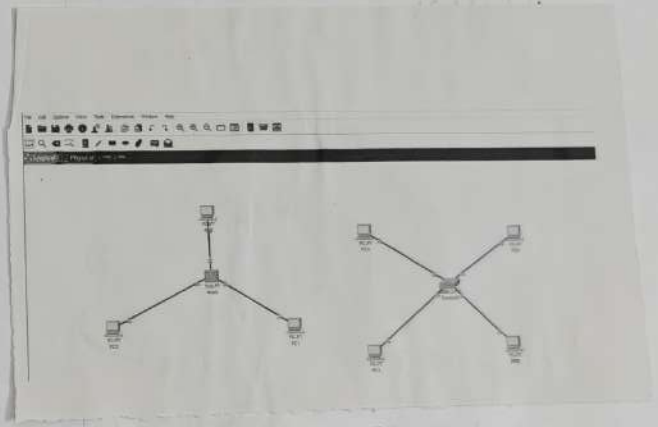
After downloading and installing wireshark, lunch it and double click the name of a network interface under capture to start capturing packets on the interface As soon as you click the interfaces name you'll see the packets start to appear in real time. Wireshark captures each packet sent to or from your system. If you have promiscuous mode enabled - it's enabled by default - you'll also see all the other packets on the network instead of only packets addressed to your network adapter. To check if promiscuous mode is enabled, click capture > options and verify the "Enable Promiscuous mode on all interfaces" check box is activated at the bottom of this window

Filtering Packets:

If you are trying to inspect something specific such as the traffic a program sends when phoning home, it helps to close down all other applications using the network so you can narrow down the traffic; still, you'll likely have a large amount of packets to shift through. That's where wireshark filter come in.







Result:

Thus the experiment, using the packet capturing tool wireshark has been successfully completed

Exp 6

HAMMING CODE

Aim: write a program to implement errors detection and correction using hamming code concept. Make a test run to input data stream and verify error correction features.

Sender Code:

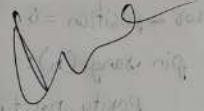
```
m = int(input("Enter the number of data bits: "))
data = input("Enter data bits: ")
if len(data) != m or any(bit not in "01" for bit in data):
    print("Invalid data!")
    exit()

r = 0
while (2**r) < (m+r+1):
    r += 1
print("Number of parity bits: ", r)
n = m+r
hamming = ['0'] * (n+1)
j = 0
for i in range(1, n+1):
    if (i & (i-1)) != 0:
        hamming[i] = data[j]
        j += 1
for p in range(r):
    parity_position = 2** p
    parity = 0
for i in range(1, n+1):
    if i & parity_position:
        parity ^= int(hamming[i])
hamming[parity_position] = str(parity)
hamming_code = ''
for i in range(n, 0, -1):
    hamming_code += hamming[i]
print("Original data: ", data)
print("Hamming Code: ", hamming_code)
```

Receiver Code:

```
received = input("Enter received Hamming Code: ")
if any(bit not in "01" for bit in received):
    print("Invalid Hamming Code!")
    exit()
n = len(received)
r = 0
while (2** r) <= n:
    r += 1
print("Number of parity bits: ", r-1)
hamming = [0] + list(received[::-1])
error_position = 0
for p in range(r):
    parity_position = 2**p
    parity = 0
    for i in range(1, n+1)
        if i & parity_position
            parity ^= int(hamming[i])
    if parity != 0:
```

```
else:
    print ["Error detected !")
    print ("Error position from Right : ", error_position)
    if hamming [error_position] == '0':
        hamming [error_position] = '1'
    else:
        hamming [error_position] = '0'
: Corrected_code = ""
: for i in range (n, 0, -1)
    Corrected_code += hamming [i]
print ("Corrected Hamming code:", Corrected_code)
```



Result: Successfully completed the experiment

Exp. 7 Flow control wrt Data link layer

Aim: Write a program to implement flow control at data link layer using sliding window protocol.

Sender Program:

```
# include <stdio.h>
# include <stdlib.h>
# Include <string.h>
# include <unistd.h>
# define MAX 100

struct frame { int frame_no;
               char data;

} int main() {
    struct frame (MAX);
    int windowsize, n, i, start=0;
    char message [MAX];
    File fp;
    int ack;
    scanf ("%d", &windowsize);
    scanf ("%s", &message);
    n=(strlen (message);
    for (i=0; i<n; i++) {
        frames [i] frame no = i+1;
        frames [i] data = message [i];
        while (start<n) {
            int end = start of windowsize;
            if (end>n)
                end=n;
```

```
    }
    else if (ack < 0) {
        ack = -ack;
    printf("NACK received for frame %d \n", ack);
        start = ack + 1;
    }
    printf("All frames transmitted successfully. \n");
    return 0;
}
```

Receiver Program :

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
int main () {
    FILE *fp, *out;
    int frameNo;
    char data;
    int expected = 1;
    while (1) {
        fp = fopen ("Sender Buffer", "r");
        if (fp == NULL) {
        printf ("Waiting");
        sleep (2);
        continue;
    }
    while (if scanf (fp, "%d %c", &frameNo, &data);
        if (frameNo == expected) {
```

```
        out = fopen ("Receiver_Buffer", "w");
        if (out == NULL) {
            fclose (fp);
            return 0;
        }
    }
    fclose (fp);
    return 0;
    }
    return 0;
}
```

Result:

The code has been executed successfully

Exp.No:08

Nmap using Try Hack Me

Aim:
To discover and understand live host on a network using Nmap host discovery scans ARP scan, ICMP scan and TCP/UDP ping scan and.

ARP Host Discovery:

$ Sudo nmap -PR -Sn 10.200.6.0/24

O/P

Starting NMAP 7.80 (https://nmap.org)
Nmap Scan report for 10.200.6.1
Host is up
Nmap done : 255 IP address (1 host up)

ICMP Host Discovery:

ICMP Echo (-PE)

$ Sudo nmap -PE -Sn 10.200.6.0/24
Nmap scan report for 10.200.6.1
Host is up (0.0698 latency)
Nmap scan report for 10.200.6.50
Host is up (0.070 latency)
Nmap scan report for 10.200.6.250
Host is up (0.069s latency)
Nmap done : 256 Ip address (3 hosts up)
Scanned in 3.38 seconds

TCP UDP Ping Discovery:

(i) TCP SXN Ping (-PS)

$ nmap -PS -Sn 10.200.6.0/24
Nmap scan report for 10.200.6.50
Host is up (0.078 latency)
Nmap scan report for 10.200.6.250

(ii) TCP ACK Ping (-PA)

Sends a TCP packet with ACK flag set
$ sudo nmap -PA -Sn 10.200.6.1/24 Nmap
Scan report for 10.200.6.50
Host is up (0.0.698 latency)
Nmap done : 256 IP address (2 hosts up) scanned
in 36.64 seconds

(iii) UDP Ping

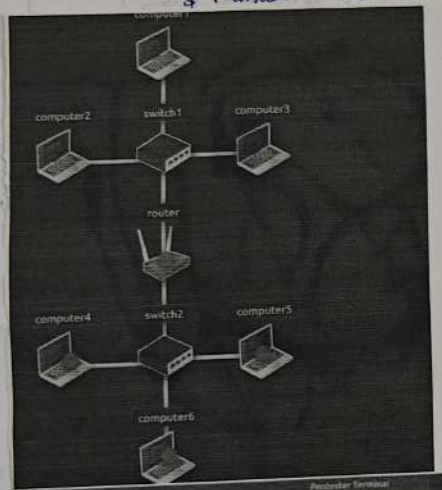
$ Sudo nmap -PU -Sn 10.200.6.0/24
Nmap scan report for 10.200.6.50
Host is up (0.0s Latency)
Nmap scan report for 10.200.6.250
Host is up (0.06) latency)
Nmap done 256 IP address (2 host up)
Scanned in 35.97 seconds

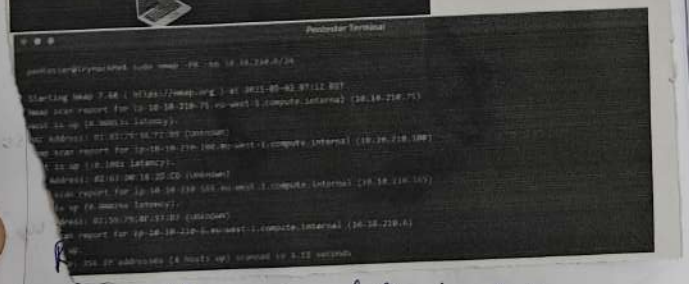
Masscan - A Faster Alternative

$ masscan 10.200.60/24 -p443

$ masscan 10.200.60/24 -p80,443

$ masscan 10.200.60/24 -p22-25





Hence successfully finished.

Expno: 09 Implementation of Subnetting in Cisco Packet Tracer

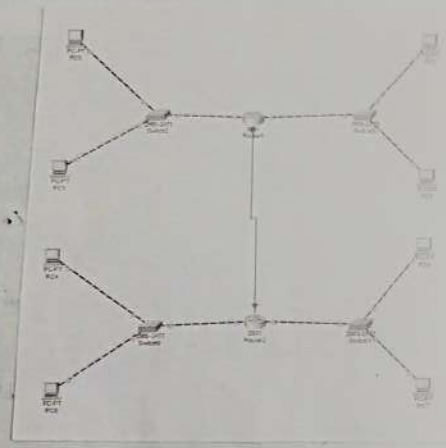
Aim:

To implement the subnetting in Cisco Packet Tracer Simulator

Procedure:

1. Open Cisco Packet Tracer and create a new network topology.
2. Add the required routers, switches and PCs to workspace.
3. Connect the device using appropriate copper cables
4. Use the network 192.168.1.0/24 and divide into /27 network
5. Configure the router interfaces with the required IP address and subnet mask 255.255.255.224
6. Use enable and config terminal in the router CLI
7. Enter the interface configuration terminal and assign IP using
   (i) ip address
   (ii) no shutdown

8. Use the ping command to test Connectivity between pc's and the router



Result:

The ping is successfull, the networks configured successfully

Inter Networking Packet

Aim:

Inter networking
PACKET TRACE

Procedure:

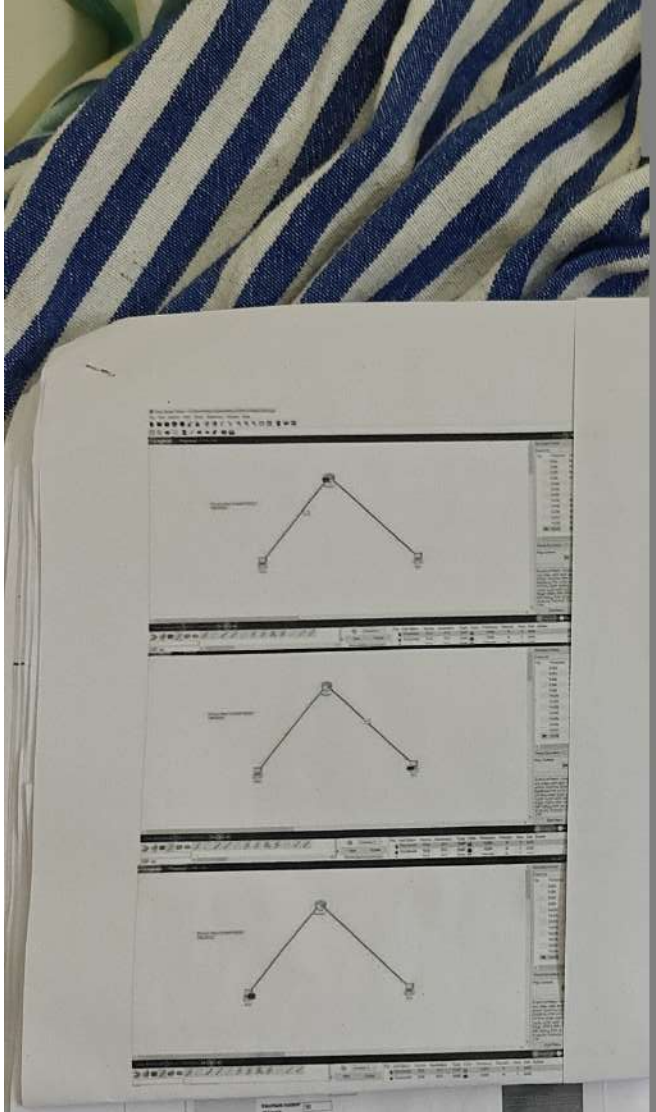
(i) Configure
(ii) Configure
(iii) Connecti

Router Config Table:-

Device name	Ip addr ethernet
Router	192.168.10.1

PC Config Table

Device name	IP Addre
PC0	192.168.10
PC1	192.168.20



Result: The internetworking with router ha
been completed successfully.

Exp. 11.a.

Simulate static routing e
CISCO Packet Trace

Aim:

To Simulate va sta
configuration using C

Procedure:

→ Open cisco packet tra
the given network

→ Connect 3 routers and
suitable cable

→ Configure IP address
and PC's

→ Enable all router
no shutdown

→ Test Connectivity

→ Disconnect the main
that the backup rou

Exp No 11.6 Simulate RIP using cisco pa

Aim :

To Simulate RIP using CISCO packet tracer.

Procedure :

→ Create the topology 3 routes and 2 PC's

→ Connect the device suitable cables

→ Assign the given I both PCs

→ Configure fast Ethe Interface of all ro

→ Enable interface usi Shutdown

→ Test Connectivity

→ Disconnect the routes 25.

Experiment :- 12 (a)

Aim :-
Implement echo client server using TCP/IP

→ Read a message from user
→ Send same to server
→ Recieve the echoed message

Experiment - 12(b)

Aim: To implement a client-client server application.

Algorithm:

I/P:

client - Hello

server - Hi

client - How are you?

server -

Experiment-13

Aim: Implement your own ping program

Program:

Experiment-14

Aim: Write a code using RAW sockets to implement packet sniffing.

Program: #include (stdio.h)

Experiment - 13

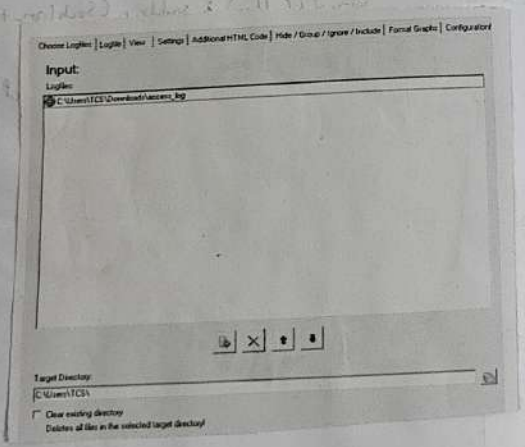
Aim:
To analyze the different types of web logs using Webalize tool.

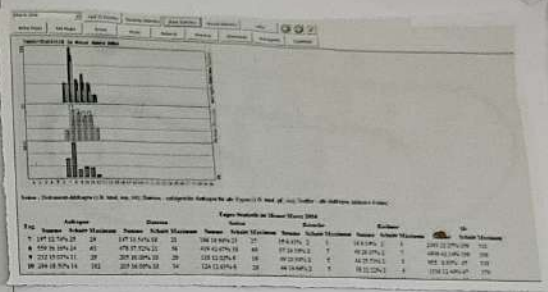
Procedure:

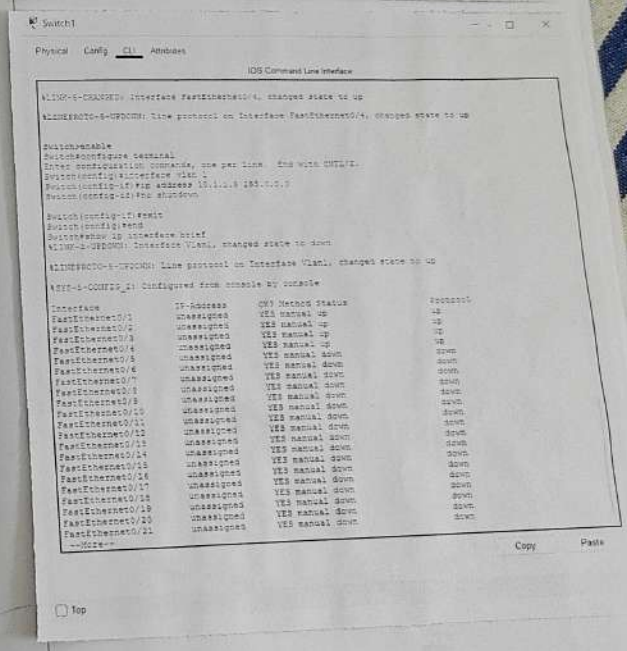
Step 1: Run Webalize win version.

Step 2: Input web log file

Step 3: Press run Webalizer







Result: The experiment has been executed successfully.